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A place to talk about Modular Synthesizers. And Cats.

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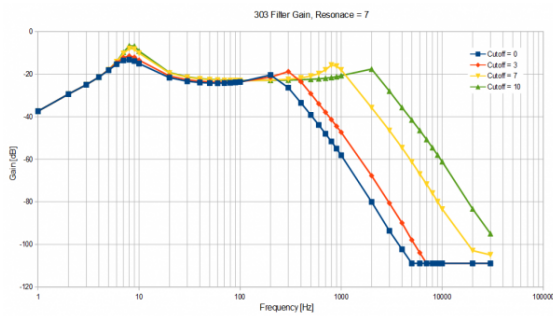
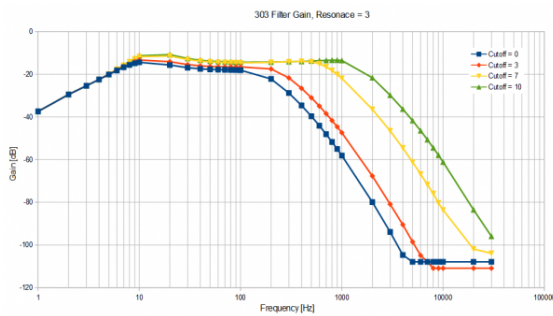
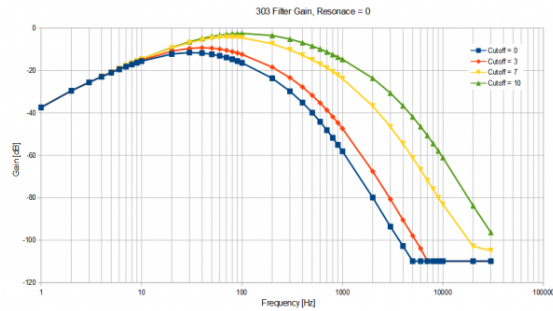
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303 circuit modelling

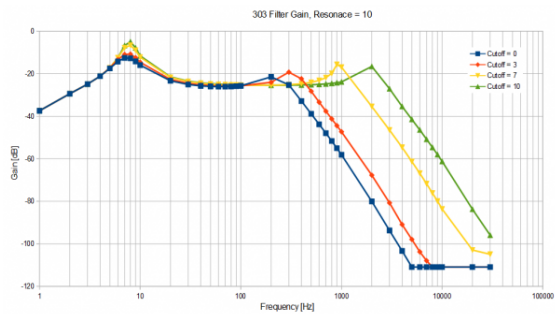
by [guest](#) » Thu Nov 21, 2024 3:30 am

i realize this is a heavily trod path, but for some reason i cant resist the urge to walk it myself. to that end, i opened up my 303 and took frequency versus gain plots for the filter. i also have VCA output plots for the same settings (with VCA help full open), but, just to get started, here is the filter output (taken from top of reso pot) with a 5Vpp sinewave input at R62. also, its worth noting that there is a decent amount of distortion in the filter for gains above -20dB.

[guest](#)
Super Deluxe Wiggler

Posts: 13314

Joined: Mon Aug 19, 2013 11:49 am



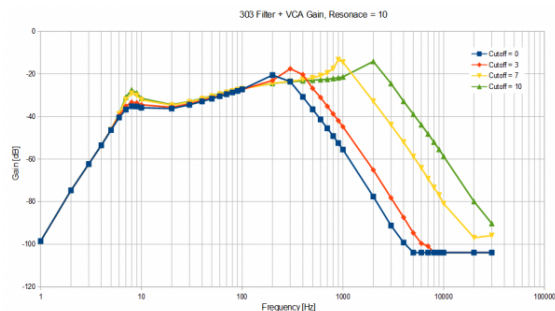
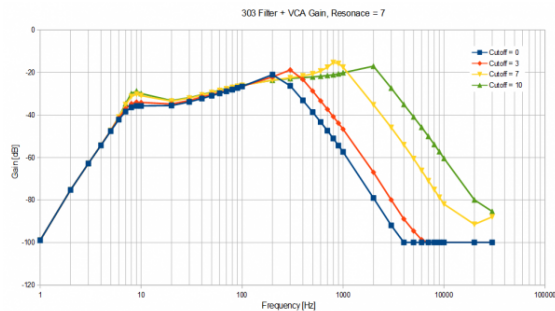
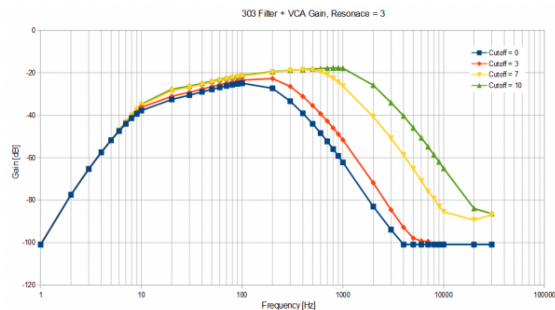
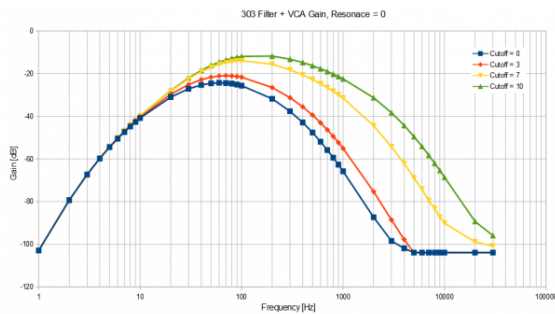
the peaks for cutoff=10 are lower than the actual resonance, as i took measurements at 2kHz, and the actual resonance is at 1.8kHz, so its 3dB or so better than whats shown. the cutoff and resonance numbers are for knob position. there are 11 ticks around the knob: 0 - 10.

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Re: 303 circuit modelling

by [guest](#) • Thu Nov 21, 2024 4:00 am

and here is the VCA output. there is a heavy highpass on the input to the VCA.



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Re: 303 circuit modelling

by [SynthfoxAstro](#) • Thu Nov 21, 2024 9:11 am

Whoa. Interesting. So does this mean a bass mod is possible by increasing the highpass capacitor? From what i seem to see in the circuit, it's a simple RC

“



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“



filter.

Aubrey 'Astro' Lis | she/her or they/them --> [SDIY website](#) / [my commercial gear](#) / [music](#) / [personal](#)

Re: 303 circuit modelling

by [guest](#) » Thu Nov 21, 2024 1:07 pm

yeah, if you change out the 2 capacitors leading to the vca, the 160Hz rolloff seen in the vca output will go away, and you will have a response as shown in the filter graphs. there is still the bass rolloff at low Q due to the diffamp at the filter output having too low of impedance. this can be fixed by reducing its bias current and increasing the collector resistors, but you get some more noise.

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Re: 303 circuit modelling

by [devinw1](#) » Thu Nov 21, 2024 2:22 pm

Interesting on that high pass. Seems a bit high for a bass machine. They were probably trying to get rid of some clicking and or thumping a bit? Would be easy enough to try say 100n caps instead of 10n in there though.

<https://www.westonaudio.com>

Re: 303 circuit modelling

by [guest](#) » Thu Nov 21, 2024 5:50 pm

yeah, it makes no sense for a bass machine, but i think its one of the things that makes it a good acid machine. the higher frequencies can be quite piercing.

i put together a mathematical model of the filter that matches quite well. the resonance is higher in the math model, but only at the very high end. there is a bit of mismatch in the subaudible region, but im not too worried about it. there is a pretty good math explanation of the filter on tim stinchcombes website:

<https://www.timstinchcombe.co.uk/index.php?page=diode>
<https://www.timstinchcombe.co.uk/index.php?page=diode2>

i did a halfway version between just the ladder and the whole circuit. it turns out that 2 of the highpass filters are doing 99% of the work in terms of determining the filter response.

$$H(s) = nks(ms+1)/((ms+1)(ks+1)((ps)^4+7(ps)^3+15(ps)^2+10ps+1)+Gmks^2)$$

p = cutoff frequency [EDIT] - (1/cutoff freq)
G = reso (0->16)
n = overall gain (1.66)
m = feedback highpass (1uF+10k) = 0.05
k = diffpair highpass (0.1uF+diffpair input impedance) = 0.02

i havent tried the frequency shifted version with the first cap being 1/2 yet, but i dont expect a lot of difference. above is done with equal caps in the theoretical filter.

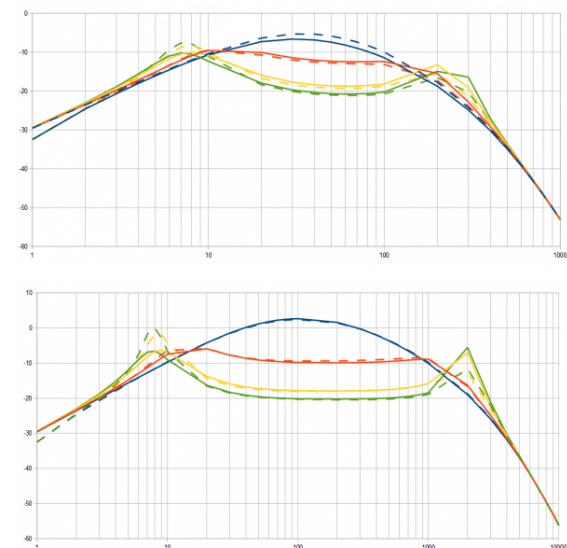
Last edited by [guest](#) on Thu Nov 21, 2024 11:57 pm, edited 1 time in total.

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Re: 303 circuit modelling

by [guest](#) » Thu Nov 21, 2024 7:25 pm

here are some graphs showing the current match i have between model and measurements. these are for cutoff at 0 and for cutoff at 10, with the resonance being varied. i used the same parameters for resonance and k,m for both of them, but the data wasnt taken at exactly the same resonance settings. i kept the cutoff frequency fixed per dataset, and varied resonance. so there are probably slight variations in the collected data with respect to resonance.



the resonance in the subaudible range doesnt go as high in the model, but this is all below 20Hz, and the lowest note the 303 produces is 32Hz. the



SynthfoxAstro
Common Wiggler

Posts: 155
Joined: Tue Nov 03, 2020 4:17 pm
Contact: [✉](#)



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



[devinw1](#)
Super Deluxe Wiggler

Posts: 4134
Joined: Tue Aug 07, 2018 11:20 am
Location: Portland, OR
Contact: [✉](#)



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

resonance at the high end is too high, and part of this is not measuring exactly at resonance, but i think there is also a compression happening at high gains with resonance. the low cutoff graph doesnt match passband amplitude all that well for the same reason, as the gain is much higher for low resonance in that regime. the filter gain also drops for low cutoff as the output impedance of the filter becomes on the same order as the input impedance of the diffamp stage that follows, so you get attenuation.

also, dashed is measured, and solid is modelled.

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Re: 303 circuit modelling

by [devinw1](#) » Fri Nov 22, 2024 1:56 pm

We shall hence forth call the feature on the response plot the "acid bump"

<https://www.westonaudio.com>

Re: 303 circuit modelling

by [guest](#) » Fri Nov 22, 2024 3:18 pm

and here is the VCA model. same as above, but with another filter added afterwards:

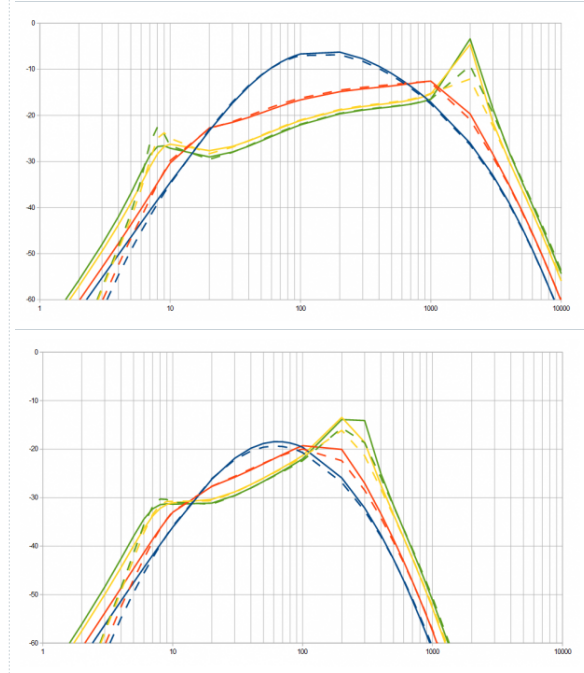
$$(as/(as+1))+(G/8)*(0.5*as/(0.5*as+1))$$

G = resonance (0->16)

a = highpass cutoff (0.013)

you can get pretty close with just $(1+G/8)*(as/(as+1))$

ATTACHMENTS



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Re: 303 circuit modelling

by [guest](#) » Fri Nov 22, 2024 10:51 pm

i tried modelling the half-cap portion of the filter using the following transfer function:

$$1/(0.5s^4+4s^3+10s^2+8s+1)$$

and it was only mildly different from the all-same-caps version i used initially. there is no difference in the lower half of the curve, and only a slight change in resonance peak, with the half-cap being smaller, and therefore better. but its so mild i cant be certain that its not due to slight variations in cutoff frequency or resonance. i also tried the transfer function for hal-cap given on stinchcombes website, and it produced identical results to my equation above.

i guess the next step is to code it up and see how it performs.

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Re: 303 circuit modelling

by [KSS](#) » Sun Nov 24, 2024 1:56 am



Re: 303 circuit modelling

by [guest](#) » Wed Nov 27, 2024 12:47 am

in the above graphs, the modelled passband amplitude is too high at low cutoff frequencies (there is a misprint in one of the graphs above, the blue lines should be swapped - dashed is above when it should be below). this is due to the way the model was put together, with each section being done independently and cascaded (filter, diffamp, feedback, etc). but, there is no buffer between the filter and the diffamp, and at low cutoff frequencies, the



[devinw1](#)
Super Deluxe Wiggler

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Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



[KSS](#)
Super Deluxe Wiggler

Posts: 15586
Joined: Mon Jan 25, 2016 7:28 am



output impedance of the filter is on the same order as the diffamp, so there is a great deal of attenuation. this chops the gain down. to model this, i treated both the diffamp and filter as one circuit, and derived the transfer function. it was quite tedious, as there are 81 terms in the full expansion, and i had to do it multiple times to get all the errors out. im pretty confident that this is right:

$$k(ps)/(0.5k(ps)^5+(4k+2)(ps)^4+(10k+13)(ps)^3+(8k+18.5)(ps)^2+(k+11)(ps)+1)$$

p = 1/cutoff frequency

k = .021/p (highpass into diffamp relative to cutoff frequency)

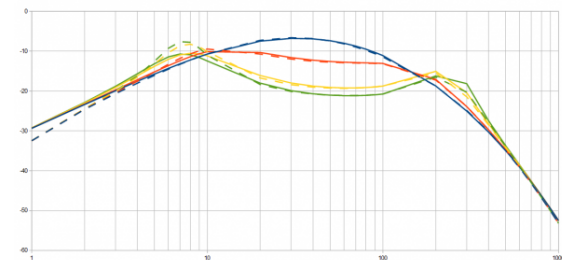
adding in the feedback and second highpass stage gives a full transfer function as:

$$(km(ps)^2+k(ps))/((m(ps)+1)(0.5k(ps)^5+(4k+2)(ps)^4+(10k+13)(ps)^3+(8k+18.5)(ps)^2+(k+11)(ps)+1)+Gmk(ps)^2)$$

G = feedback (0->16)

m = .045/p (highpass in feedback relative to cutoff frequency)

here is the lowest cutoff frequency with the new model, and it matches pretty well. it matches better the higher the frequency, but those worked pretty well with the old model, so i didnt bother posting it. the resonance is tamped by 3dB or so as well. but that issue can really only be tackled by including the tanh distortion, which i cant model as easily with the way im currently doing it.

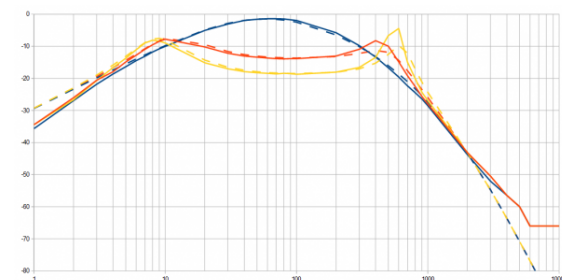


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Re: 303 circuit modelling

by [guest](#) • Sat Nov 30, 2024 1:00 am

ok, some progress, finally. i spent a bunch of time writing algorithms, and none of them would operate very fast. i want to run this on a microcontroller, and preferably one thats not just a mini-computer. the calculating of the IIR filter tap constants is pretty ridiculous, and i wanted to do that at a pretty fast rate (2ksps or better). the IIR filter i came up with was 12 taps, which is reasonable, but there is other thing the micro will have to do besides for the filter, so it didnt make me super confident. then i realized that the filter tap constants could all just be pre-computed. even if i went with 1k filter settings (10 cent resolution), thats only 14kB of storage for 16b constants. then i got to wondering how bad it would be to just use an accumulator per capacitor and just make a digital replica of the analog circuit directly, with multiplies for the resistors. this got me down to 6 multiplies and i could run it on an 8b arduino (which i happened to have at home):



this is data taken from an arduino uno with codec shield: function generator in and scope at the output to measure frequency response. im actually quite pleased with this. the dashed line is my mathematical model (which is shown above to match to the real pretty well). i didnt use the actual data as it was difficult to tune to exactly the same cutoff frequency, which the math model could easily do. there is a really high noise floor for a few reasons, the biggest is that its 16b fixed point math, with a lot of multiplies and accumulation (only 16b accumulators as well), so the noise picks up and the true rolloff cant be seen.

the trick here, is that the actual circuit can be mimicked with the wrong capacitor values in the code. if you have a series of (RCs+1) that get multiplied together, you can tune the resultant polynomial by choosing different values of RC for each of the poles. so even though my RC circuit is an idealized, buffered system, i can fake the nonideal, unbuffered 303 filter by moving the poles around. again, this is something that will have to be precomputed, but its also super low computation at runtime. i tuned these via trial and error, but if i decide to go this route, i can tune them via computer to match the math model. in theory the ratios should be fixed, but i dont think its going to play out that way as the output impedance of the ladder filter changes with frequency. this is another advantage to the precomputed constants - it simplifies that impedance problem as well.

one final thing - after spending the day collecting data, i kept wondering why the arduino was matching the 303 so well at the super low end (<5Hz), and when i was all done, i realized i had the scope set to AC couple, which i also had done when measuring the 303. so all of that low end roll off is the scope. ill characterize the rolloff the scope and then add it back in to the collected data, just in case its helpful to someone in the future. i plan on posting the spreadsheet with all the collected data and math models when im done.

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Re: 303 circuit modelling

by [kimballa](#) • Sat Nov 30, 2024 2:50 pm

Guest, that is some very cool modeling! Congratulations on nailing down all that detail.

I am wondering about the need for the math runtime optimization though. Cycle-optimizing and minimizing the fixed point multiplication is a neat hack... But do you think optimizing / targeting for the 8 bit AVR is worth it? For basically the same price as a baseline Arduino (\$20-25 range) you can get an ATSAM51- or STM32-powered one, which will do 100 MHz+ and has a hardware floating point coprocessor. And if you would then integrate the mcu into a board directly, they're only about \$5/ea in qty 1, also not much more expensive than an AVR which runs about \$3.50.

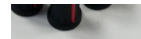
As a bonus there's a lot more FLASH ROM too, so if you do want to precompute stuff, there's a lot more room to spare. And since the hardware is 32 bit, you could use 32 bit floating point math and lower that noise floor by another several dB.

"For the challenge" is a totally valid answer, as is "I have more AVR chips in my house than I know what to do with." 😊 I'm just curious. There's plenty of ways I'm making my work harder on myself by virtue of shunning digital components entirely for analog ones in many projects, so I get it. But at least for me when I "do" go digital, a slightly higher bom cost for a considerably more-powerful MCU just opens up so much more capability as well as the ability to just relax and think about the functionality when I'm coding rather than be locked in battle with the limitations of the architecture. (This may also stem from the fact that I have a handful of ATSAM51 Arduinos sitting around waiting for a project to grab them...)

Re: 303 circuit modelling

by [guest](#) • Sat Nov 30, 2024 4:30 pm

totally fair question, and one i ask myself regularly. also, you reminded me i have an adafruit feather here, which should be a lot faster (but doesnt have a codec). i think ive also convinced myself that there is no way this will ever run on an 8b machine without being super lofi, although thats also part of the fun, seeing if it can be done. right now im doing 6 multiplies and ive used up all my time (although i dont know why, i feel like it should be closer to 20 multiplies per sample period).



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

[kimballa](#)
Wiggling with Experience

Posts: 277
Joined: Sun Jan 29, 2023 4:04 pm
Contact: [📧](#)



from a more practical perspective, i think there are 2 or 3 reasons why im trying to reduce computation:

1. this is mostly a learning project for me, as i typically dont do a lot of digital. ive written a few filters before, but they were either super simple (CIC or rolling average), or i used an online FIR calculator. so reducing it makes it easier for me to understand and implement all the steps along the way.
2. im thinking of making a little pocket-x0x, with sequencer, and id like it to be battery powered, so getting it to run on an MO would be great.
3. id like to write it in arduino, so other folks can play with it as well, and the more platforms it can run on (not needing a teensy, for example), would increase the access to it.

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Re: 303 circuit modelling

by [guest](#) » Sun Dec 01, 2024 8:40 pm

it seems like the most common way of going from a transfer function to a digital filter is via the bilinear transform. this produces an IIR filter with $2n-1$ multiplies, where n is the order of the transfer function (eg $1/(s^2+s+1)$ has $n=2$). it also has a very large range of coefficients. for the filter im working on, some coefficients are 400x to 4000x others, which means floating point or 32b multiplies are required. thats not necessarily a problem, but it does mean a fair bit of computation. in contrast, the simple $1/(s+1)$ filter can be implemented with a single multiply, and the coefficients are much closer together. so to make a 5th order filter, you can cascade these:

$$es/(as+1)(bs+1)(cs+1)(ds+1)(es+1)$$

this is a 4th order lowpass and a first order highpass. but, its restricted in terms of what the filter response can be. the filter i want to make has the transfer function:

$$ks/(0.5ks^5+(4k+2)js^4+(10k+13)js^3+(8k+18.5)js^2+(k+11)js+1)$$

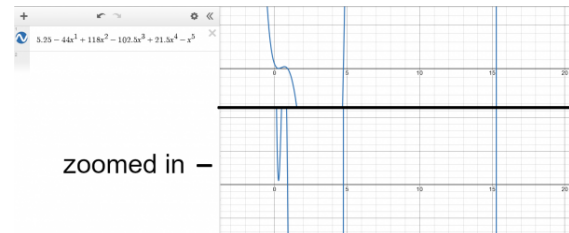
with $k=0.021^*f$, where f is the cutoff frequency. so for any cutoff setting, the k can be translated to fixed numbers. one such example, for a cutoff frequency around 500Hz, is:

$$1/(5.25s^5+44s^4+118s^3+102.5s^2+21.5s+1)$$

so the difficult part becomes, what values of a,b,c,d,e will produce the coefficients i need (5.25, 44, etc) when the first equation above is expanded out $(abcde)s^5 + \dots$? the full expansion has symmetry in the coefficients (all of the inputs are the same form), so you can solve for any one of the variables (a,b,c etc) and get all of the answers at once. the form of the answer is:

$$k5-k4a+k3a^2-k2a^3+k1a^4-a^5=0$$

where $k5$ is the input s^5 coefficient (5.25 in our case), and a is the output coefficient were solving for. i have plotted this equation in the graph below.



the lower plot is a zoomed in version of the lower plot, just to show that it doesnt cross zero at 0.28. what we want are the zero crossings, the places where the equation is zero, this gives 0.9, 4.75, and 15.28. but, there are supposed to be 5 coefficients, so whats going on? it turns out that you cant perfectly match for all cases, and this one is a bit off. if you take the next closest-to-zero point of 0.28, and you use that point twice, you get a really good approximation. im not sure exactly how you know which point to use twice, or which one i supposed to be the highpass coefficient, but i knew what it was supposed to look like, and which coefficients were close, so i was able to obtain:

5.12, 43.68, 117.7, 102.41, and 21.49 for my coefficients, which is a pretty close match to the original:
5.25, 44.00, 118.0, 102.50. and 21.5

matching at the low end was more important to me than the high end, so those coefficients ($k1, k2, k3$) match more closely. i was going to post a graph of both the theoretical and approximation, but the two lines perfectly overlap and it looks like one curve, so not sure it actually shows anything.

at any rate, maybe this is of use to someone, or maybe its just a fun read. im pretty happy to have stumbled upon this, as i think it makes my work a bit easier.

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Re: 303 circuit modelling

by [toneburst](#) » Mon Dec 02, 2024 10:07 am

This is cool to see, though way over my head technically, sadly.

Apologies if this isn't the right place for this, but it did get me thinking. I once tried to create a 303 emulator in Supercollider, but didn't get very close, sadly.

A couple of questions I never found an answer to, and maybe someone here can help

- Does the 303 filter cutoff track oscillator pitch at all?
- What's are the approximate shapes of the filter envelope and lag/slide curves?

Keep up the great work guys!

Re: 303 circuit modelling

by [guest](#) » Mon Dec 02, 2024 12:32 pm

thanks for following along!

1. the 303 filter does not track pitch (at least stock doesnt, you can mod for this).
2. there are 2 envelopes, one for vca, one for vcf, and they are both just attack/decay. the attack is around 3ms, and the decay is variable for the vcf, and fixed at 6s or so for the vca. the decays are exponential (RC curves), with a few slight oddities to them. for the vca, the envelope is gated by the gate, so for fast notes, you dont hear much of the RC at all, and its just on/off. its slides where you hear the full envelope. accents are a different story. you get a low-passed version of the envelope going to both the vca and vcf, each with different time constants. the vcf lowpass also has a diode in it, so the cv build over time as successive notes are played.

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Re: 303 circuit modelling

by [kimballa](#) » Mon Dec 02, 2024 9:02 pm

[guest](#) wrote:

Sat Nov 30, 2024 4:30 pm
totally fair question, and one i ask myself regularly. also, you reminded me i have an adafruit feather here, which should be a lot faster (but doesnt have a codec).

(...)

from a more practical perspective, i think there are 2 or 3 reasons why im trying to reduce computation:



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Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

[toneburst](#)
Super Deluxe Wiggler

Posts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London



[guest](#)
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

[kimballa](#)
Wiggling with Experience

Posts: 277
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1. this is mostly a learning project for me, as i typically dont do a lot of digital. ive written a few filters before, but they were either super simple (CIC or rolling average), or i used an online FIR calculator. so reducing it makes it easier for me to understand and implement all the steps along the way.
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3. id like to write it in arduino, so other folks can play with it as well, and the more platforms it can run on (not needing a teensy, for example), would increase the access to it.

Those are all pretty good reasons! Makes sense 😊

I love the Adafruit Feather M4 Express and have used it in a bunch of projects. I think it's one of the best bang for buck Arduinos, and the form factor is super compact. The ATSAM51 has a lot of great peripherals in it. The 12 bit 2-channel ADC is actually surprisingly good, and runs at 1MHz. Since you only need 44 kHz for audio, that means you can implement 16x oversampling (with some headroom to spare) and get 14 ENOB precision.

I haven't used the DAC on the ATSAM51 myself yet, but it also has a two-channel, 12 bit, 1 MSPS DAC peripheral with DMA support. Might get you where you need to go. If you would have an ADC input channel free, you could wire the DAC back to the ADC - oversampling and reading at 14 bits - and use that for closed loop control. If you cut a trace (at an SMD jumper pad on the pcb) that hardwires the pin to 3V3, you can also use an external low-noise / low-ppm AREF like REF35250 for better repeatability.

Re: 303 circuit modelling

by **toneburst** » Tue Dec 03, 2024 4:37 am

“ guest wrote: 📎

thanks for following along!

Mon Dec 02, 2024 12:32 pm

1. the 303 filter does not track pitch (at least stock doesnt, you can mod for this).
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Great, thank you very much for that! I'll look at modifying my emulation, with all that in mind.

Re: 303 circuit modelling

by **toneburst** » Tue Dec 03, 2024 4:45 am

“ guest wrote: 📎

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Mon Dec 02, 2024 12:32 pm

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Great, thank you very much for that! I'll look at modifying my emulation, with all that in mind.

“ guest wrote: 📎

accents are a different story. you get a low-passed version of the envelope going to both the vca and vcf

Mon Dec 02, 2024 12:32 pm

So I guess that means effectively that the peak of the envelope is smoothed off and the attack increases on accented notes? But also, the VCF envelope decay is much shorter (and a fixed length), isn't it?

Re: 303 circuit modelling

by **guest** » Tue Dec 03, 2024 11:58 am

yeah, thats probably the main thing, the vcf decay time is forced to its shortest setting. the attack stays the same, though, and the vca envelope stays the same. the lowpass filtering of the vca envelope into the vca for accents is a very subtle effect, and mostly just increases the attack a bit (which is maybe the attack you were mentioning above?), but its just a millisecond or so. the time constant of the filter is 500us.

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Re: 303 circuit modelling

by **guest** » Tue Dec 03, 2024 12:10 pm

“ kimballa wrote: 📎

I love the Adafruit Feather M4 Express and have used it in a bunch of projects. I think it's one of the best bang for buck Arduinos, and the form factor is super compact. The ATSAM51 has a lot of great peripherals in it. The 12 bit 2-channel ADC is actually surprisingly good, and runs at 1MHz. Since you only need 44 kHz for audio, that means you can implement 16x oversampling (with some headroom to spare) and get 14 ENOB precision.

Mon Dec 02, 2024 9:02 pm

thanks for all the suggestions. ive used the m4 express once before (or maybe it was the m0?), and it was a bit confusing to get set up (required special board files), but once i got it going, i was impressed with what it could do. i liked that it had compiler settings so i could make the code run faster. i think it may have even unrolled loops. it also has a battery charger chip onboard.

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Re: 303 circuit modelling

by **toneburst** » Wed Dec 04, 2024 8:20 am

“ guest wrote: 📎

the lowpass filtering of the vca envelope into the vca for accents is a very subtle effect, and mostly just increases the attack a bit

Tue Dec 03, 2024 11:58 am

Cool, thanks for the clarification.

Re: 303 circuit modelling

by **guest** » Fri Dec 06, 2024 3:13 pm

ok, so ive moved on to the square wave for a short bit. after the filter, the square is probably the most unique aspect of the 303. it changes its shape and pwm as you change frequency (and there is some slew rate to this effect). the higher the note, the lower the duty cycle. i originally wrote this code for an 8b arduino which was running at a lower clock rate, so the digital time constants are off by 10% or so, but its got the right shapes. this is running on feather M4 express, and takes about 2us to compute (around the same amount of time the filter takes). it will need some tuning when im satisfied that its glitch free, but all phases are tunable. the way the square wave is generated is via a single transistor being used as a comparator, with an RC circuit computing the time average for the comparison point. this RC filter is what causes the pwm to shift, as the charge time changes with frequency. this model computes the RC comparison point and transistor in-out characteristics for 4 regions: $V_{be} < 0$ (cutoff), $0 < V_{be} < 0.6$ (exponential), $V_{be} > 0.6$ (linear active), $V_{ce} = 0$ (saturation). there is some wobble, which is probably truncation of the capacitor voltage, but i should be able to fix that. heres a video of it in action:

“ **toneburst**
Super Deluxe Wiggler

Posts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London

“ **toneburst**
Super Deluxe Wiggler

Posts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London



“ **guest**
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am



“ **guest**
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“ **toneburst**
Super Deluxe Wiggler

Posts: 1042
Joined: Fri May 02, 2014 2:41 pm
Location: London



“ **guest**
Super Deluxe Wiggler

Posts: 13314
Joined: Mon Aug 19, 2013 11:49 am

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Re: 303 circuit modellingby **guest** » Sat Dec 07, 2024 2:15 pm

just a quick update: i tried an implementation of the full, 5 pole transfer function using the bilinear transform, and it would not stabilize. it pretty much immediately blew up. i could get it to sort of work using double precision floats on the computer, but single precision on the microcontroller was a no go. so my options are implementing it as a series of 2nd order biquads, or what i originally did, which is a series of first order systems. i decided on the first order systems, as i think that will inherently be more stable, mimics the original circuit topology, and takes fewer computations. i think it will have bad frequency matching at higher frequencies (cutoff frequency not where it should be), but i can compensate for that as im using lookup tables for the coefficients. the downside is that the gain of the stages need to be modulated with cutoff frequency, which i was worried would end up at the output as AM, but a friend suggested modulating the input to compensate, and i think that should work.

i have a mockup of this running on the M4F, both in floating point and 16b ints. the 16b int version is incredibly stable, in that it doesnt blow up, ever, but it has a lot of warble and modulation as it dithers around those last bits. the floating point version is pristine (seriously, i didnt think i would get anything looking this good), but it can go unfixably stable at too high of resonance. but, if i put limits on the resonance, that shouldnt happen. i might also be able to saturate my floats. the int version takes 2.4us to run, whereas the float takes 2.9us, which is crazy to me. ultimately, im going to write 2 different versions of the emulator when im done: one that runs on floats, and the other that uses a mix of 16b and 32b ints, so that it can be used on all sorts of platforms.

finally, i was banging my head against the wall for 2 hours last night trying to figure out why the code wasnt working. it turns out an int in arduino is 32b if compiling for ARM, and 16b if compiling for AVR. what a nightmare. i usually define things explicitly as uint16_t, but was being lazy and it bit me. im so used to writing optimizations for the 8b AVR, as it cant do that much, that im actually writing really bad code for these ARM platforms. not only does it not work at times (depending on int rollover, etc), but its just not any faster and is less readable and portable.

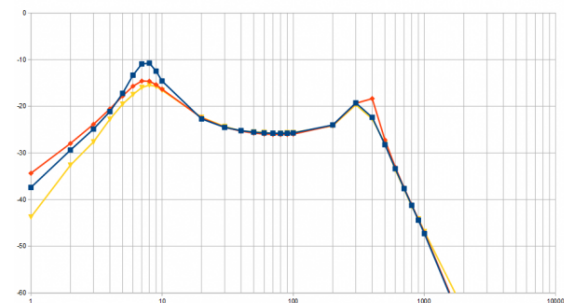
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Re: 303 circuit modellingby **guest** » Sun Dec 08, 2024 10:23 pm

ok, here are some results from the latest filter. this is floating point on an M4F, using cascaded single poles, and implementing a tanh distortion for the filter core. this last bit is the critical bit. it tamps down the distortion to match the actual device. below is a plot of three things:

1. blue - actual 303
2. red - full math model
3. yellow - data taken from microcontroller

the math model always pushed the resoance higher, both in terms of frequency and amplitude. with the tanh model, the resonance now matches the actual 303 perfectly. there is some issue at the low end where the microcontroller doesnt match its frequency exactly to the math model. i am not sure why this is. i could manually tweak it to match, but i dont think its an issue as there are no frequencies of interest down there, and it doesnt match the actual 303 anyways. i didnt bother modelling all of the really low frequency highpass filters, so the resonance is off way down there.



this is for full resonance. i didnt bother posting the lower resonance graphs as all the lines are on top of each other anyways. im going to run a test at the highest frequency as well, as this is both where resonance errors were the worst, and where frequency warping will be a larger issue in the digital filter. but for now i am extremely happy with this.

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Re: 303 circuit modellingby **Bjarne** » Mon Dec 09, 2024 12:41 am

This thread is just great - both educational and entertaining! I just love the transparent log book approach of threads like this.

Re: 303 circuit modellingby **guest** » Mon Dec 09, 2024 10:38 pm

ok, so some more things fixed today. i feel like im pretty close to getting some reasonable sound out of this thing. i was facing 2 issues: 1. the high frequency rolloff was not right for high cutoff frequencies, and 2. the resonance at high cutoff frequencies would go unstable. the first was an error in how i was calculating the lowpass coefficients. i just assumed they were T/w , where T is the sample period, and w is the cutoff frequency, but as you get closer to the sampling frequency, the relative time gets shorter. so i did a first order compensation for this by using $T/(w+T/2)$, so the higher frequency coefficients get pushed down a bit. there is a second order compensation for this that can be done, but so far i dont think i need to do that, as i dont need that kind of accuracy.

“

**guest**
Super Deluxe WigglerPosts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“

**guest**
Super Deluxe WigglerPosts: 13314
Joined: Mon Aug 19, 2013 11:49 am

“

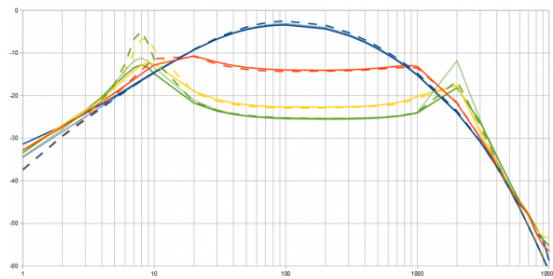
**Bjarne**
Common WigglerPosts: 187
Joined: Sat Jul 20, 2024 10:31 am
Location: Stockholm

“

**guest**

the second issue i fixed by changing the order i calculated my filter in. because there is feedback, there is a choice to be made on how you process the feedback. do you add it first and calculate the filter, or do you calculate the filter and add it last? i was doing the latter, and i think that was increasing the delay in the feedback loop by a sample, and therefore causing early self-resonance.

the final thing i figured out was that a lot of my calculations were from the atmega, which had a 44.1ksp sample rate, and now im using 50ksp, which is why the low frequency bump was in the wrong place. so its looking much better:



the long dashed lines are the 303, and the solid are the microcontroller. the fine dashed lines are the math model. i left those in for the high resonance case to show how much the tanh helps (perhaps a bit too much, i might back it down a bit) and that the no resonance case doesnt match the 303, but does match the model. im not sure why the model is off there, but it is minor and im not going to worry about it at this point.

and one final "im an idiot" anecdote: up until today, i had been taking data by playing the samples out the onboard dac, measuring on a scope, and then manually entering them into a spreadsheet. but the data is already just numbers inside of the microcontroller, so now im just having the data spit out on the serial console, and im copy-pasting into the spreadsheet. so much easier, oh my god!

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Re: 303 circuit modelling

by [guest](#) » Sun Dec 15, 2024 2:57 pm

so, i got my first audio out, and its ok. it took a long time to generate the lookup tables for the cutoff, and im still not done with the bottom portion of the table. in general, i think the filter has the right overall sound, but the resonance feels a bit weak. the output is from a 12b, onboard dac, with a really bad powersupply, so there is a lot of noise. sorry about that. im going to need to get a decent codec on this sooner or later. this is just the filter being driven with open vca and an arpeggiated sawtooth. there are some odd artifacts, and i think some of them are due to the resampling when sending to my computer, others might be aliasing. first pass is full resonance and second pass is no resonance.

0:00 / 0:00

303_test1.mp3

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